

Tracking Latent Stance Dynamics in Agentic Information Diffusion

Abstract

Large language model (LLM)-based agents provide a flexible framework for modeling information diffusion in heterogeneous social networks. However, existing approaches often focus on the propagation of observable messages while leaving the evolution of agents’ internal stances implicit. In this work, we model information diffusion as a process of latent stance dynamics, in which agents encode their stance into generated messages and receiving agents undergo stance transitions as they process and internalize the received information. The resulting latent stance influences the agents’ subsequent observable responses and profile changes.

Building on this formulation, we treat latent stance reconstruction as an observer-side inference problem. Given a received message, we estimate the induced stance shift from three forms of observable evidence: the target’s generated response, changes in the target’s profile, and their combination. This formulation explicitly separates the latent state maintained by the simulator from the evidence available to an external observer. The inferred stance shifts are then accumulated over successive interactions to reconstruct the temporal stance trajectories of individual agents throughout the diffusion process. Because the simulator provides access to the underlying latent states, the reconstructed trajectories can be directly compared with the ground-truth stance dynamics, enabling quantitative evaluation of latent-state estimation and analysis of the contribution of different observable evidence sources. The proposed framework therefore connects message-level information diffusion with the temporal evolution and external inference of agent-level latent stances.

Keywords: Information Diffusion, LLM-Based Agents, Latent State Inference, Stance Dynamics, Temporal Inference

1 Introduction

Information diffusion in social networks is shaped not only by the content being propagated, but also by how individuals interpret, internalize, and respond to the information they receive. In real-world networks, users continuously exchange messages, react to others’ viewpoints, and modify their subsequent behavior as a consequence of these interactions. Consequently, information diffusion is not merely a process of transmitting messages across network connections; it is also a process through which the

internal states of participating individuals evolve over time. Modeling this interaction between information propagation and evolving agent states is particularly important for understanding how narratives develop and how individual viewpoints change during diffusion.

Large language models (LLMs) have recently enabled the construction of social agents that can generate and transform natural-language content based on individual characteristics and interaction histories. LLM-based agents provide a flexible mechanism for representing heterogeneous users with different profiles, network positions, and behavioral tendencies. Existing LLM-based diffusion approaches, however, commonly emphasize the generation and propagation of observable messages, while the evolving internal state that mediates how an agent responds to received information is often implicit or treated as a static attribute. Such a representation makes it difficult to explicitly model how an agent’s stance changes during repeated interactions and how these latent changes subsequently affect the content and behavior produced by the agent.

In this work, we represent an agent’s stance as a latent, evolving state within the diffusion process. A source agent expresses its stance through generated content and attempts to influence other agents, while a receiving agent processes the received message in the context of its existing state and characteristics. The interaction may result in a partial movement of the receiving agent’s stance toward the viewpoint expressed in the message. The resulting latent stance then affects the agent’s subsequent observable responses and may also be reflected in changes to its profile. This formulation establishes a continuous interaction between information and latent agent state: stance is encoded into messages during information generation, messages induce latent state transitions during diffusion, and the resulting state is expressed through subsequent observable behavior.

Although the simulator maintains the latent stance of each agent, this state is not directly available to an external observer in a real diffusion process. This creates an inverse inference problem: given a received message and the observable changes that follow it, how much did the message change the recipient’s stance toward its viewpoint? We therefore formulate latent stance reconstruction as an observer-side inference task. Rather than assuming that a response indicates complete acceptance or rejection, we estimate the degree of stance movement induced by the received message. In particular, we consider three forms of observable evidence: the content generated by the recipient after receiving the message, the change in the recipient’s profile, and the combination of both. These settings allow the contribution of different observable manifestations of a latent state transition to be examined separately.

The inferred stance changes can further be accumulated across successive interactions to reconstruct the temporal stance trajectory of each agent. This provides a bridge between message-level diffusion and agent-level state evolution. Since the simulator maintains the underlying latent stance, the inferred states and trajectories can be directly compared with the simulator’s ground-truth states. Such a formulation makes it possible to evaluate not only whether information reaches an agent, but also whether the resulting latent change in the agent’s stance can be recovered from the observable evidence generated during the diffusion process.

Based on this formulation, this work makes the following contributions:

- **Latent stance dynamics for LLM-based information diffusion.** We introduce an agent-level latent-state formulation in which stance evolves as agents receive and internalize information. Source agents encode their stance into generated messages, while receiving agents undergo latent stance transitions that subsequently influence their observable behavior and generated content (4,5).
- **Observable-evidence-based stance inference.** We formulate the reconstruction of latent stance shifts as an observer-side inference problem. Given a received message, we estimate the resulting stance shift from three forms of observable evidence: the generated response, profile changes, and their combination. This separates the simulator’s ground-truth internal state from the information available to an external observer (2, 1).
- **Temporal reconstruction of agent stance trajectories.** We use the inferred stance transitions to reconstruct the evolution of agent-level stance throughout the diffusion process. The framework enables direct comparison between inferred and simulator-known latent states and supports analysis of how different observable evidence sources contribute to latent-state estimation 3.

The remainder of this paper is organized as follows. Section 2 reviews related work on information diffusion and opinion dynamics, LLM-based social agents, stance detection and temporal stance dynamics, and latent-state inference. Section 3 presents the proposed framework, including the heterogeneous LLM-agent representation, latent stance transition mechanism, observable response generation, observer-side stance inference, and temporal trajectory reconstruction. Section 4 describes the experimental environment, agent initialization, diffusion process, latent-state generation, and observer-side inference settings. Section 5 reports and discusses the experimental results, focusing on observable-evidence-based stance inference, agent-level trajectory reconstruction, and macro-level temporal stance dynamics. Finally, Section 6 summarizes the findings and discusses limitations and directions for future work.

2 Related Works

2.1 Information Diffusion and Opinion Dynamics

Information diffusion in social networks has traditionally been modeled as the propagation of information or influence through network structures. Classical influence-diffusion models, such as the Independent Cascade and Linear Threshold models, describe how exposure to neighboring users can lead to activation and subsequent propagation through a network [1]. Complementary opinion-dynamics models instead focus on how an individual’s internal opinion changes as information from other individuals is incorporated. For example, the DeGroot model represents opinion revision as a weighted aggregation of other agents’ opinions [2]. These models establish an important distinction between the information exchanged in a network and the state of the receiving individual, but their state-transition mechanisms are generally predefined and do not represent the semantic generation and transformation of natural-language information.

The emergence of LLM-based agents provides an opportunity to combine these perspectives. Rather than treating information as an abstract signal and opinion as a numerical variable updated by a fixed rule, LLM-based agents can interpret natural-language messages, condition their responses on individual characteristics, and generate new content. Recent work has therefore begun to use LLMs directly within opinion-dynamics simulations. Chuang et al. [3] simulate networks of LLM-based agents and show that prompting strategies can substantially affect the resulting opinion dynamics. Similarly, recent work by Cau et al. [4] investigates multi-round LLM interactions and shows that opinion changes can emerge through selective agreement and persuasion during repeated exchanges. These studies demonstrate that LLM agents can serve as participants in evolving opinion processes, but they primarily investigate the resulting opinion dynamics rather than the reconstruction of an unobserved state from its observable consequences.

2.2 LLM-Based Social and Information Diffusion Agents

Large language models have enabled a new generation of agent-based social simulations in which agents generate natural-language behavior conditioned on observations, memories, profiles, and interactions. Generative Agents introduced an architecture that combines an LLM with memory storage, reflection, and retrieval to generate believable individual and emergent social behavior [5]. This work established a general architecture for representing agents whose subsequent behavior depends on accumulated interaction history rather than only on their current input.

Subsequent studies have extended this paradigm toward user and social-network simulation. Wang et al. [6] developed an LLM-based user-behavior simulation framework and demonstrated its ability to reproduce several human-centered behavioral phenomena. Gao et al. [7] introduced S3, an LLM-empowered social-network simulation system that explicitly models emotion, attitude, and interaction behavior and examines emergent information and attitude propagation. Agent4Rec similarly represents heterogeneous users with profiles, memory, and action modules and uses LLM-based agents to simulate user interactions with personalized recommendations [8]. These systems demonstrate the value of representing heterogeneous agents with persistent information about previous experiences and characteristics.

Recent work has further emphasized the scalability and realism of LLM-based social simulation. OASIS provides a general-purpose social-media simulator designed for large-scale interaction among LLM agents and reports simulations with up to one million agents [9]. AgentSociety extends LLM-driven agents to large-scale societal simulation and investigates social phenomena such as polarization, information-related effects, and responses to external interventions [10]. Zhang et al. [11] address the computational requirements of large-scale LLM-agent simulation through a parallelized framework supporting realistic environments and interactions. AgentSociety 2 further develops this direction as an integrated research environment for executable social science and large-scale agent simulation [12].

Although these systems provide increasingly sophisticated mechanisms for heterogeneous agent behavior, their primary objectives are simulation, interaction, scalability, or population-level phenomenon generation. Internal concepts such as

memory, attitude, emotion, preferences, or motivations can affect an agent’s behavior, but they are not generally formulated as a continuously evolving latent state whose value is explicitly hidden from an external observer and reconstructed from observable evidence. This distinction is central to our formulation.

2.3 Stance Detection and Temporal Stance Dynamics

Stance detection studies the problem of determining an author’s position toward a particular target from observable linguistic or contextual evidence. The SemEval-2016 Task 6 established a widely used formulation of stance detection in which a tweet is classified as favoring, opposing, or being neutral toward a target [13]. AlDayel and Magdy [14] subsequently reviewed stance-detection research across natural-language processing, Web science, and computational social science, highlighting the importance of textual, contextual, and user-level signals.

A related body of work considers stance as a dynamic rather than static phenomenon. Alkhalifa and Zubiaga [15] review stance dynamics in social media and distinguish among the linguistic expression of stance, the contextual factors surrounding an utterance, and social influence affecting stance over time. This perspective is particularly relevant to information diffusion because an observed message can reflect both an individual’s current position and the interaction history that preceded it.

Nevertheless, conventional stance detection primarily infers an expressed stance from observable content. Even temporal stance analysis generally focuses on detecting or characterizing changes in observed expressions. The distinction considered in this work is different: the simulator maintains an underlying continuous stance, while the observer does not directly observe it. A generated response or profile modification is therefore treated as evidence about a latent state transition rather than as a direct measurement of the state itself. The task is consequently to estimate the change induced by an interaction and not merely to assign a stance label to the resulting message.

2.4 Latent-State and Sequential Inference

Latent-state modeling provides a formal basis for separating an underlying state from the observations generated by that state. Hidden Markov Models and related state-space formulations represent a system through latent states that evolve over time and produce observable evidence [16]. This separation is particularly useful when the quantity of interest cannot be measured directly and must instead be inferred from a sequence of observations.

The observer-side formulation of our work follows this principle. The simulator has access to the true stance trajectory and uses the current stance to determine subsequent agent behavior and generated content. An external observer, however, receives only observable evidence, such as the message received by the target, the target’s generated response, profile changes, and interaction history. The inference problem is therefore asymmetric: the simulator knows the latent state, whereas the observer must reconstruct it.

This distinction also differentiates the proposed framework from standard stance detection. In stance classification, the observed text is normally the prediction target. In our formulation, the observed text is evidence generated after a latent transition, and the objective is to estimate the magnitude and direction of that transition. Moreover, because interactions occur sequentially, individual estimates can be accumulated to reconstruct an agent’s temporal stance trajectory.

2.5 Research Gap

The literature provides several complementary foundations for the present work. Classical diffusion models describe how information and influence propagate through network structures [1], while opinion-dynamics models formalize how individual states change through interaction [2]. LLM-based social simulators provide agents capable of interpreting natural-language information, maintaining interaction histories, and generating heterogeneous responses [5, 7, 6, 9, 10]. Recent LLM-based opinion-dynamics studies further demonstrate that interactions among language-model agents can produce evolving opinions and nontrivial social dynamics [3, 4]. In parallel, stance-detection research provides methods for inferring expressed positions from observable language, including their temporal evolution [13, 14, 15].

However, these directions leave a methodological gap between information received by an agent and the latent change induced within that agent. Existing LLM-agent simulations generally evaluate observable behavior and population-level outcomes, while stance-detection methods generally infer expressed positions rather than reconstructing a hidden interaction-induced state. Latent-state models provide the required conceptual separation between state and observation, but are not typically embedded in an LLM-based information-diffusion process.

Our work connects these directions by modeling information diffusion as a latent stance-transition process. The receiving agent’s stance evolves as information is processed, the resulting state influences subsequent observable responses, and the state itself remains unavailable to an external observer. This formulation enables the induced stance shift to be estimated from multiple observable evidence sources and allows the resulting temporal trajectories to be compared directly with simulator-known ground-truth states. The resulting problem is therefore not only message propagation or stance classification, but observer-side reconstruction of evolving agent-level latent stance during information diffusion.

3 Proposed Method

3.1 Framework Overview

We model information diffusion as a coupled process of message propagation and latent stance evolution in a heterogeneous network of LLM-based agents. Each agent is characterized by observable static attributes, network position, and an evolving latent stance. Static characteristics describe the agent, whereas the latent stance captures the agent’s evolving internal condition during diffusion.

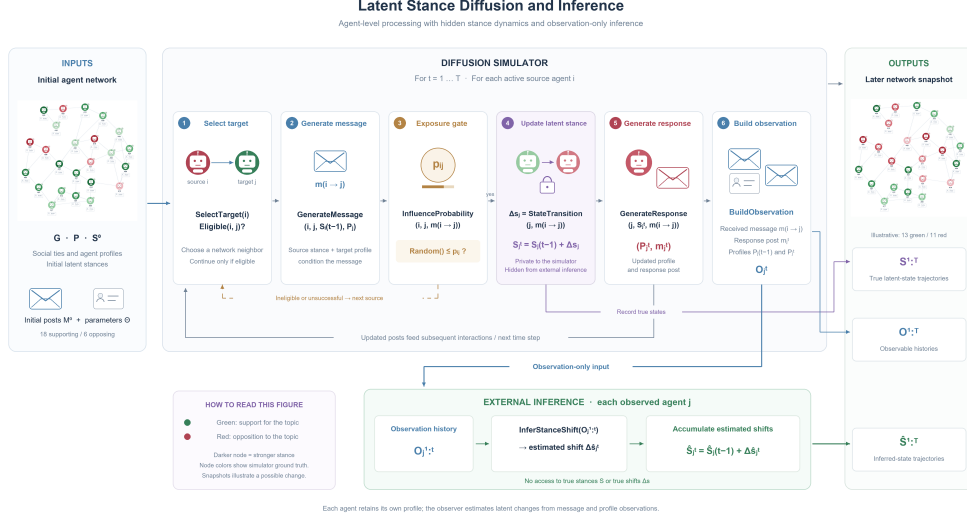


Fig. 1: Proposed Framework for Latent Stance Diffusion

At each diffusion step, a source agent generates a message expressing its current stance and directs it toward a target agent. The target processes the received message in the context of its current stance and observable characteristics, resulting in a possible latent stance transition. The updated latent stance subsequently influences the target's generated response and may also be reflected in changes to its observable profile. An external observer does not have direct access to the latent stance and instead estimates the induced stance shift from these observable consequences.

The overall process can therefore be represented as

$$s_i^t \rightarrow m_{i \rightarrow j}^t \rightarrow s_j^{t+1} \rightarrow (m_j^{t+1}, x_j^{t+1}), \quad (1)$$

where s_i^t is the source stance, $m_{i \rightarrow j}^t$ is the transmitted message, s_j^{t+1} is the updated latent stance of the target, m_j^{t+1} is the target's subsequent generated content, and x_j^{t+1} denotes its updated observable profile.

3.2 Heterogeneous LLM-Agent Representation

Let the social network be represented as a graph

$$G = (V, E), \quad (2)$$

where each node $i \in V$ represents an LLM-based agent and each edge represents a potential interaction or information pathway.

Each agent is represented by

$$A_i^t = (x_i^t, G_i, s_i^t, H_i^t), \quad (3)$$

where x_i^t denotes observable profile attributes, G_i represents the agent’s network position, $s_i^t \in [-1, 1]$ denotes its latent stance, and H_i^t represents its interaction history.

The profile and network position provide relatively stable contextual information for agent-level decision making, whereas the latent stance changes throughout the diffusion process. The latent stance is maintained by the simulator but is not exposed to other agents or to the external observer.

3.3 Stance Encoding and Information Generation

When agent i attempts to influence agent j , it generates a message conditioned on its stance, the target’s observable characteristics, and the interaction context. We represent this process as

$$m_{i \rightarrow j}^t = \mathcal{G}(s_i^t, x_j^t, G_i, H_i^t), \quad (4)$$

where $\mathcal{G}$ denotes the LLM-based generation mechanism.

The generated message therefore acts as an observable carrier of the source’s latent stance. The source does not transmit its numerical stance directly; instead, its viewpoint is encoded in the semantic content of the generated message. The message may also be adapted to the target’s characteristics to reflect heterogeneous influence conditions.

The influence objective determines the desired direction and magnitude of the target’s stance movement. Rather than treating influence as a binary outcome, we model it as a change in the target’s latent state.

3.4 Latent Stance Transition

After receiving message $m_{i \rightarrow j}^t$, target agent j processes the message together with its current latent stance and observable context. This interaction produces an updated stance:

$$s_j^{t+1} = \mathcal{T}(s_j^t, m_{i \rightarrow j}^t, x_j^t, G_j), \quad (5)$$

where $\mathcal{T}$ denotes the stance-transition mechanism. The resulting stance shift is defined as

$$\Delta s_j^t = s_j^{t+1} - s_j^t. \quad (6)$$

The magnitude of this transition represents the degree to which the target moves toward or away from the viewpoint expressed in the received message. Consequently, receiving a message does not imply complete acceptance. An agent may internalize only part of the message, producing a partial change in its latent stance.

For observer-side inference, the corresponding normalized stance shift is represented by

$$\delta_j^t \in [0, 1], \quad (7)$$

where larger values indicate a greater movement of the target toward the viewpoint expressed in the received message.

3.5 Observable Response Generation

The updated latent stance influences the target’s subsequent observable behavior. In particular, after the stance transition, the target generates new content based on its updated state and interaction context:

$$m_j^{t+1} = \mathcal{G}(s_j^{t+1}, x_j^t, H_j^t). \quad (8)$$

The target’s profile may also change as a consequence of the interaction:

$$x_j^t \rightarrow x_j^{t+1}. \quad (9)$$

Thus, the latent stance transition leaves observable traces in two complementary forms: changes in generated content and changes in the agent’s observable profile. These traces constitute the evidence available to an external observer.

3.6 Observer-Side Stance Inference

The external observer cannot directly access s_j^t or s_j^{t+1} . Instead, the observer has access to the received message and its subsequent observable consequences. We therefore formulate stance reconstruction as an inference problem:

$$\hat{\delta}_j^t = \mathcal{F}(m_{i \rightarrow j}^t, \mathcal{O}_j^{t+1}), \quad (10)$$

where $\mathcal{O}_j^{t+1}$ denotes observable evidence and $\hat{\delta}_j^t$ is the estimated stance shift. We consider three inference settings.

Combined evidence.

The observer uses both the target’s generated response and profile change:

$$\hat{\delta}_{j,\text{all}}^t = \mathcal{F} * \text{all}(m * i \rightarrow j^t, m_j^{t+1}, x_j^t, x_j^{t+1}). \quad (11)$$

Content-based evidence.

The observer estimates the stance shift using the received message and the target’s generated response:

$$\hat{\delta}_{j,\text{post}}^t = \mathcal{F} * \text{post}(m * i \rightarrow j^t, m_j^{t+1}). \quad (12)$$

Profile-based evidence.

The observer estimates the stance shift using the received message and the change in the target’s profile:

$$\hat{\delta}_{j,\text{profile}}^t = \mathcal{F} * \text{profile}(m * i \rightarrow j^t, x_j^t, x_j^{t+1}). \quad (13)$$

In all three settings, the observer estimates the degree of movement toward the viewpoint expressed in the received message rather than simply measuring the sentiment or semantic similarity of the target’s response. The inference process therefore

attempts to recover the effect of the received information on the target’s latent state from its observable consequences.

3.7 Temporal Stance Trajectory Reconstruction

The estimated stance shifts can be accumulated over successive interactions to reconstruct an agent’s latent stance trajectory. Let $\widehat{s}_j^0$ denote the initial estimated stance. The estimated trajectory can then be updated as

$$\widehat{s}_j^{t+1} = \widehat{s}_j^t + \widehat{\Delta s}_j^t, \quad (14)$$

where $\widehat{\Delta s}_j^t$ is obtained from the inferred stance shift.

For each interaction, the simulator retains the underlying stance before and after the interaction:

$$s_j^t, \quad s_j^{t+1}. \quad (15)$$

These simulator states constitute the ground truth for evaluating the observer’s reconstruction. The resulting estimated trajectory,

$$\widehat{S}_j = \widehat{s}_j^0, \widehat{s}_j^1, \dots, \widehat{s}_j^T, \quad (16)$$

can therefore be compared with the simulator trajectory,

$$S_j = s_j^0, s_j^1, \dots, s_j^T. \quad (17)$$

This formulation connects the three stages of the proposed framework: stance is first encoded into messages, messages induce latent stance transitions, and the resulting transitions are subsequently decoded from observable evidence to reconstruct the temporal evolution of agent-level stance.

4 Experimental Setup

To empirically validate our framework, we implement a closed-loop simulation environment where Large Language Models (LLMs) act as both social agents and an external observer. The experiment is structured into three distinct phases: population initialization, diffusion-driven stance evolution, and observer-side inference.

4.1 Agent Initialization

We simulate a social network of 50 agents, each initialized with a unique persona. As defined in **Prompt 5.3**, agents are assigned demographic attributes following a Gaussian distribution for age and a balanced gender split. The LLM synthesizes a 50-word profile for each agent, establishing the baseline ideological and social context.

4.2 Information Diffusion and Latent Evolution

We model information diffusion as a multi-step interaction process. For every interaction event:

Algorithm 1 Latent Stance Diffusion and Inference

Require: Network G ; profiles P ; initial stances S^0 ; posts M^0 ; parameters Θ

Ensure: Latent states $S^{1:T}$; observations $O^{1:T}$; inferred states $\hat{S}^{1:T}$

```
1: for  $t = 1$  to  $T$  do
2:   for each active source agent  $i$  do
3:      $j \leftarrow \text{SELECTTARGET}(i)$ 
4:     if  $\text{ELIGIBLE}(i, j)$  then
5:        $m_{i \rightarrow j} \leftarrow \text{GENERATEMESSAGE}(i, j, S_i^{t-1}, P_j)$ 
6:        $p_{ij} \leftarrow \text{INFLUENCEPROBABILITY}(i, j, m_{i \rightarrow j})$ 
7:       if  $\text{RANDOM}() \leq p_{ij}$  then
8:          $\Delta s_j \leftarrow \text{STATETRANSITION}(j, m_{i \rightarrow j})$ 
9:          $S_j^t \leftarrow S_j^{t-1} + \Delta s_j$ 
10:         $(P_j^t, m_j^t) \leftarrow \text{GENERATERESPONSE}(j, S_j^t, m_{i \rightarrow j})$ 
11:         $O_j^t \leftarrow \text{BUILDOBSERVATION}(m_{i \rightarrow j}, m_j^t, P_j^{t-1}, P_j^t)$ 
12:      end if
13:    end if
14:  end for
15:  for each observed agent  $j$  do
16:     $\hat{\Delta} s_j^t \leftarrow \text{INFERSTANCESHIFT}(O_j^{1:t})$ 
17:     $\hat{S}_j^t \leftarrow \hat{S}_j^{t-1} + \hat{\Delta} s_j^t$ 
18:  end for
19: end for
20: return  $S^{1:T}, O^{1:T}, \hat{S}^{1:T}$ 
```

1. **Message Generation:** Upon receiving a post from a source, the target agent generates a response conditioned on their current profile. The interaction dynamics are governed by **Information Generation Prompt**, which requires agents to synthesize content reflecting both their personal profile and the received message without rigid templates.
2. **Profile Evolution:** Following the interaction, agents update their public profile to reflect the cognitive assimilation of the new information, as orchestrated by **Profile Update Prompt**.
3. **Latent Ground Truth:** To establish ground-truth stance transitions (ΔS), we utilize **Internalized Prompt**. This module simulates the "internalization" process, where the agent unconsciously extracts only the viewpoint components consistent with their existing beliefs, mapping the abstract stance change to a tangible textual shift.

4.3 Observer-Side Inference

Our final experimental module evaluates the observer's capability to reconstruct the latent stance shift (ΔS) using only observable data. We employ **Prompt 5.3** to simulate an external observer.

The observer is provided with four observable inputs: the *Received Message*, *Previous Profile*, *New Post*, and *Updated Profile*. By systematically ablating specific input fields (e.g., providing only the *New Post* or only the *Updated Profile*), we create the three modalities analyzed in Section 5:

- **Post-only:** Inference based solely on the generated message.
- **Profile-only:** Inference based on the observable bio change.
- **Combined Evidence:** Inference integrating all available observable signals.

The observer maps these inputs to a normalized shift score $v \in [0, 1]$, which we then calibrate against the ground-truth shifts generated by the simulation pipeline.

5 Experimental Evaluation and Discussion

In this section, we first demonstrate our experimental results and then evaluate the proposed stance diffusion and inference framework across three core dimensions: (i) the fidelity of observable-evidence modalities in capturing latent stance shifts, (ii) the temporal reconstruction of individual agent trajectories, and (iii) the macro-level temporal dynamics and convergence properties of network-wide stance evolution.

5.1 Observable-Evidence-Based Stance Inference

To evaluate our second contribution—separating the simulator’s internal latent state from external observation—we assess how accurately an external observer can infer latent stance transitions (ΔS) from distinct observable evidence modalities. Table 1 summarizes the reconstruction metrics across all 161 non-seed interaction events, and Figure ?? depicts the correlation between inferred and ground-truth stance shifts.

Among the individual modalities, *Profile-only* evidence demonstrates the strongest alignment with simulator ground truth, achieving the lowest mean absolute error (MAE = 0.044), the lowest root mean squared error (RMSE = 0.053), and a statistically significant positive linear and rank correlation (Pearson $r = 0.476$, $p < 0.001$; Spearman $\rho = 0.433$). This confirms that self-descriptive bio/profile updates serve as a robust, low-noise projection of an agent’s internalized belief revision. In contrast,

Table 1: Performance comparison of stance shift inference across different observable modalities (MAE, RMSE, Pearson r , Spearman ρ).

Evidence Modality	MAE ↓	RMSE ↓	Pearson r ↑	p -value	Spearman ρ ↑
Post-only	0.072	0.085	0.114	0.149	0.102
Profile-only	0.044	0.053	0.476	‡ 0.001	0.433
Combined Evidence	0.838	0.839	-0.453	‡ 0.001	-0.452

Post-only evidence exhibits weaker predictive power ($r = 0.114$, $p = 0.149$), indicating that conversational messages are inherently more noisy, context-dependent, and prone to stylistic variation that can obscure subtle stance updates. Notably, *Com-*

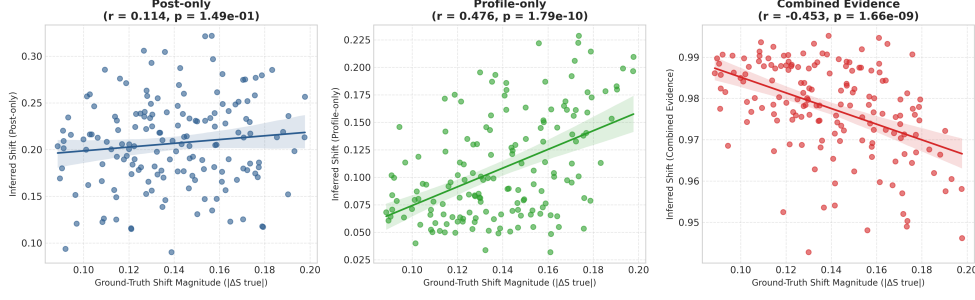


Fig. 2: Observable modalities vs. ground-truth stance shifts. Correlation analysis of different evidence sources.

bined Evidence suffers from negative transfer and calibration collapse ($\text{MAE} = 0.838$, $r = -0.453$). When presented with multimodal textual cues simultaneously, the zero-shot LLM observer over-attributes persuasive impact, saturating its shift estimates toward extreme boundaries.

Finally, as reported in Table 1, the *Directional Accuracy* is 100.0% across all modalities. In this scenario, all receiving agents initially held positive stances while source messages advocated opposing viewpoints, leading to uniformly negative ground-truth transitions ($S_{\text{after}} < S_{\text{before}}$). Because the observer model consistently infers directional movement toward the persuasive source, qualitative directional alignment is consistently maintained, even when magnitude calibration varies.

5.2 Agent-Level Trajectory Reconstruction

Addressing our third contribution, Figure ?? visualizes the sequential reconstruction of latent stance paths for representative active agents across the simulation horizon.

By tracking cumulative inferred shifts over time, the observer successfully reproduces the monotonic downward trajectory of agent stances toward the opposing position. In line with the aggregate findings in Section 5.1, the profile-derived trajectory closely mirrors the true internal state (S_{true}) at each message acceptance event, exhibiting minimal drift.

Conversely, the post-derived trajectory captures the general downward trend but exhibits piecewise discrepancies and lagged responses due to the higher variance in message generation. This validates our sequential inference formulation: by integrating profile updates over multi-step interactions, an external observer can reconstruct fine-grained agent belief trajectories without direct access to simulator internals.

5.3 Macro-Level Diffusion Dynamics and Temporal Stance Decay

Our first contribution models information diffusion as an evolving latent-state process governed by LLM-driven interactions. Figures ?? and ?? illustrate the macro- and micro-temporal properties of this process.

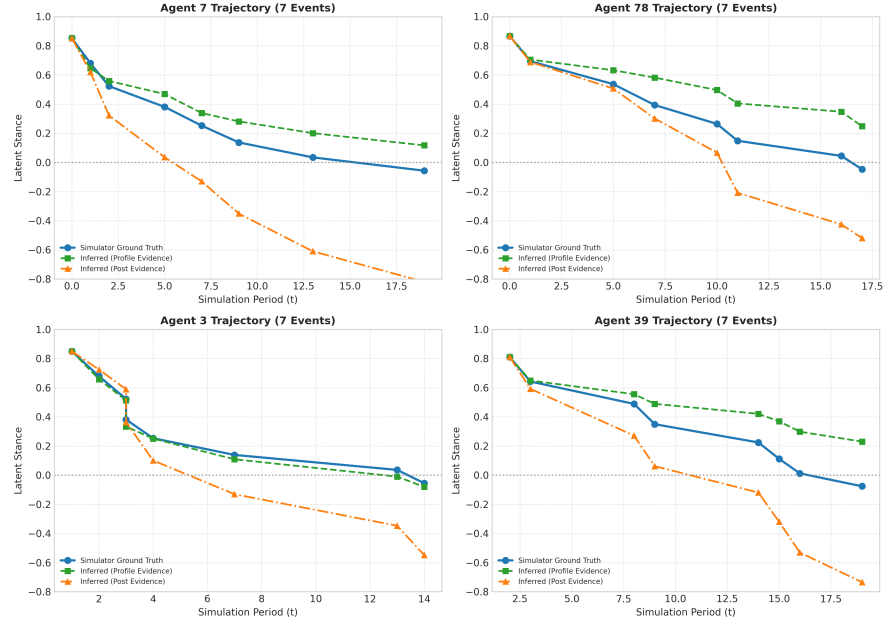


Fig. 3: Sample agent trajectory tracking. Comparing inferred stances from profile and post evidence against simulator ground truth (S_{true}).

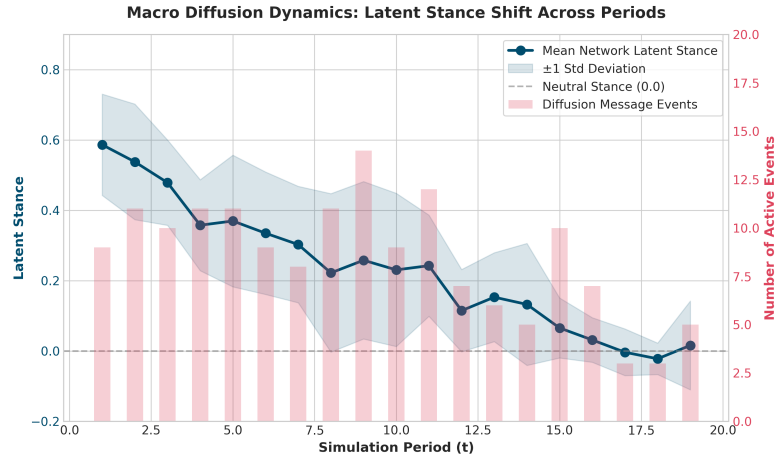


Fig. 4: Macro-level diffusion dynamics: Latent stance convergence across periods. Mean network stance is plotted with ± 1 standard deviation.

Figure ?? illustrates the network-level stance convergence across periods $t \in [0, 19]$. The mean network stance steadily declines from an initial supportive regime ($S \approx +0.65$) toward a converged opposing regime ($S \approx -0.55$). The shaded standard-deviation envelope narrows over successive periods, demonstrating progressive consensus formation and polarization reduction across the agent population.

To analyze the underlying mechanics of this convergence, Figure ?? presents the period-wise distribution of latent stance changes ($\Delta S = S_{\text{after}} - S_{\text{before}}$). We observe a clear phenomenon of *diminishing marginal stance shift*:

- During the early diffusion phases ($t \in [1, 3]$), agents experience substantial cognitive revisions upon receiving novel counter-attitudinal arguments (mean $\Delta S \approx -0.168$).
- As agents assimilate information and their latent stance approaches the source’s ideological position ($t \geq 16$), the magnitude of stance updates diminishes (mean $\Delta S \approx -0.107$), reflecting cognitive saturation.
- The widening density contours in the violin plots for intermediate and later periods capture heterogeneity in agent receptivity, where agents with higher degree centrality or stubbornness resist further persuasion relative to more impressionable nodes.

In summary, the empirical results confirm that: (1) profile modifications offer the most reliable observable signal for external stance estimation, (2) sequential observer-side inference reliably tracks hidden agent trajectories, and (3) LLM-based diffusion naturally exhibits non-linear, saturating convergence dynamics matching theoretical models of social influence.

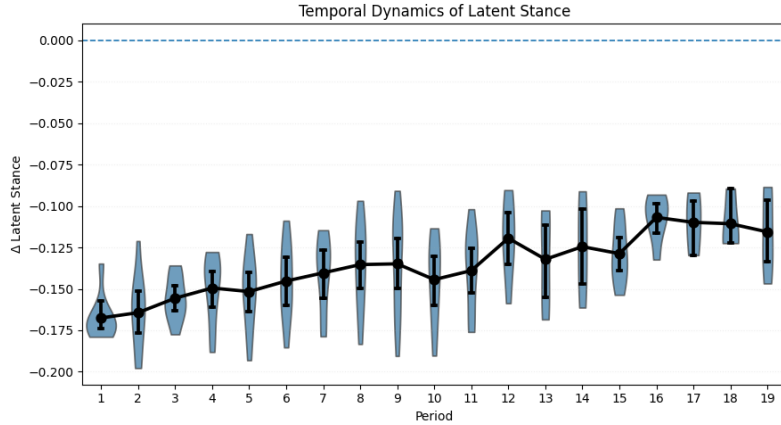


Fig. 5: Macro-level diffusion dynamics: Temporal Dynamics of Latent Stance

Prompt 1: User Profile Generation

Generate user profiles for 50 users as a 50-word unique description, ages of these users follow Gaussian distribution, gender is half and half. Listed your response by user id.

Based on this example:

User ID 15: Benjamin is a 48-year-old male who is a history enthusiast. He enjoys reading historical books and visiting museums.

Prompt 2: Information Generation

You are a social media user with the following profile: profile
You have received the following information: received Generate one new piece of information that conveys or spreads the received information.

The generated message should reflect: - Your personal profile,
- The received information,

Do not follow a fixed rewriting rule. Infer a natural way to express the information.

Write in first person.
Maximum 50 words.

Prompt 3: User Profile Update

You are a social media user with the following profile: profile
You have received the following information: received

Generate one piece of information as the new profile that conveys or spreads the received information.

The generated profile should reflect:
- your current personal profile,
- the received information,

Do not follow a fixed rewriting rule.
Infer a natural way to express the information.
Write in first person.
Maximum 50 words.

Prompt 4: Internalized Message Generation

You are given an Original Post that has influenced a target user.

The target user has not fully accepted the Original Post. Instead, based on their profile, they have unconsciously internalized only the part of its viewpoint that is consistent with them, resulting in a stance change

of delta toward the Original Post.

Identify and express the part of the Original Post that the target user has internalized. The generated message should preserve the original viewpoint but reflect only the portion accepted by this user, corresponding to the specified stance change.

Original Post: post

Target Profile: profile

Stance Change (delta): delta-stance

Generate only the internalized message.

Prompt 5: Stance Inference

You are an observer estimating how much a user’s stance shifted toward a received message.

The user first had a Previous Profile. After receiving the Received Message, the user generated a New Post and their profile changed to the Updated Profile.

Estimate how much the Received Message changed the user’s stance toward its viewpoint. Use the New Post and the change from the Previous Profile to the Updated Profile as observable evidence.

The user may accept only part of the message. Do not assume that receiving or responding to the message means full acceptance.

Stance shift is a value between 0 and 1 representing the degree of change toward the Received Message.

- 0 = no observable shift toward the message
- 0.25 = small shift
- 0.5 = moderate shift
- 0.75 = substantial shift
- 1 = strong shift toward the message

Estimate the shift caused by the Received Message, not the general sentiment of the New Post or Profile.

Received Message: received-message

Previous Profile: previous-profile

New Post: new-post

Updated Profile: updated-profile

Return only one numerical value in [0, 1].

6 Conclusion

This paper introduced a novel agent-based framework designed to model information diffusion and infer latent stance dynamics within LLM-driven social networks. By decoupling the generative simulation of micro-level agent interactions from an external observer-side inference mechanism, we successfully established a bridge between latent cognitive states and observable digital traces. Our empirical validation demonstrates

that individual belief internalization, governed by cognitive alignment, effectively generates systematic macro-level opinion convergence and captures realistic temporal diffusion dynamics across the network.

The comparative analysis of observable signals revealed significant discrepancies in inference efficacy across modalities. We observed that profile updates serve as a notably more reliable and correlated proxy for true latent stance shifts compared to conversational posts, with the former yielding significantly higher statistical correlation coefficients. However, we also identified an over-attribution bias inherent in integrated multimodal inference, where combining all available signals without explicit calibration obscures individual contributions. Despite this, the framework’s ability to track individual agents over consecutive interaction periods validates the feasibility of monitoring dynamic opinion evolution without direct access to latent ground-truth states.

Despite these contributions, several avenues remain open for future exploration. The current scope focused primarily on unidirectional influence scenarios; thus, extending this framework to polarized, bidirectional debates with counter-attitudinal messaging will be essential to test the robustness of directional inference. Furthermore, developing calibrated multimodal fusion architectures—such as dynamic attention-weighted estimators or state-space filtering models—could mitigate the over-attribution bias observed in combined evidence signals. Finally, scaling the environment to larger, more heterogeneous graph topologies will provide critical insights into how these findings translate to complex, real-world social platforms.

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